

Unsupervised Geometry & Dynamics for Anomaly Detection

Stochastic Delay Embeddings, Dual (L, d) Selection, & Anomaly Codimension

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1. Anomaly Scoring & The Codimension Objective

Time-Delay Coordinate Map:

$$\mathbf{x}_t = [x_t, x_{t+1}, \dots, x_{t+L-1}]^\top \in \mathbb{R}^L$$

Tangent Space via Carré du Champ Operator SVD:

$$\Gamma(\mathbf{x}_t) = V \Sigma V^\top \implies V_d = [v_1, \dots, v_d] \in \mathbb{R}^{L \times d}, \quad V_d^\top V_d = I_d$$

Orthogonal Anomaly Projection (Flow Velocity at $t = 0.95$):

$$s(\mathbf{x}_t) = \|(I - V_d V_d^\top) v_\theta(\mathbf{x}_t, t = 0.95)\|_2$$

Codimension $k = L - d$ Tension:

$$\lim_{L \rightarrow d} k = 0 \implies \text{Subspace Starvation (No Orthogonal Space)}$$

$$\lim_{L \gg L^*} k \gg 1 \implies \text{Ambient Noise Dilution}$$

Objective: Find (L^*, d^*) maximising off-manifold anomaly separation.

2. Unsupervised Tuning: Synthetic Anomaly Generation

1. Ambient Noise & Manifold Validation Window:

$$\mathbf{x}_t \in \mathcal{D}_{\text{val}} \subset \mathbb{R}^L, \quad \boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, I_L)$$

2. Pure Orthogonal Injection ($\mathbf{u}_\perp \in \ker(V_d^\top)$):

$$P_\perp = I_L - V_d V_d^\top, \quad \mathbf{u}_\perp = \sigma_{\text{scale}} \cdot \frac{P_\perp \boldsymbol{\xi}}{\|P_\perp \boldsymbol{\xi}\|_2} \implies V_d^\top \mathbf{u}_\perp = \mathbf{0}$$

where $\sigma_{\text{scale}} = \kappa \cdot \sigma_{\text{local}} \cdot \sqrt{L}$ preserves scale across dimensions.

3. Synthetic Anomaly Construction & Restoring Drift:

$$\mathbf{x}_{\text{anom}} = \mathbf{x}_t + \mathbf{u}_\perp \implies \text{Restoring force: } v_\theta(\mathbf{x}_{\text{anom}}, t = 0.95)$$

4. Off-Manifold Contrast Ratio (CR):

$$\text{CR}(L, d) = \frac{\mathbb{E}_{\mathbf{x}_t \in \mathcal{D}_{\text{val}}} \|P_\perp v_\theta(\mathbf{x}_{\text{anom}}, t = 0.95)\|_2}{\mathbb{E}_{\mathbf{x}_t \in \mathcal{D}_{\text{val}}} \|P_\perp v_\theta(\mathbf{x}_t, t = 0.95)\|_2}$$

3. Unsupervised Selection Loss: Codimension & Complexity

Unified Selection Loss Objective:

$$\mathcal{S}(L, d) = \underbrace{\frac{1}{1 + \mathcal{J}_{\text{norm}}(L, d)}}_{\text{Term 1: Discriminative Power}} \times \underbrace{\left(1 + \beta \frac{L}{N_{\text{tr}}}\right)}_{\text{Term 2: Complexity Penalty}}$$

Term 1: Codimension-Normalized Objective $\mathcal{J}_{\text{norm}} = (\text{CR}^2 - 1) \cdot (L - d)$

- $(\text{CR}^2 - 1)$: Baseline variance subtraction per degree of freedom (excess SNR).
- $(L - d) = \dim(\mathcal{N}_x \mathcal{M})$: Orthogonal degrees of freedom; penalizes subspace starvation ($d \rightarrow L$).

Term 2: Sample Complexity Penalty ($\beta = 0.05$)

- N_{tr} : Length of training partition ($M = N_{\text{tr}} - L + 1$ embedded windows).
- L/N_{tr} : Fraction of series consumed per window; penalizes rank degeneracy.

Invariance: Multiplicative product provides scale invariance across diverse datasets.

4. The dharm Experiment: Physical Regime Stratification

1. Rhythmic Limit-Cycle Oscillators ($N = 207$ Series):

- **Domains:** Bio-medical ECG (MITDB, SVDB, LTDB), motion (OPP), SED, Power.
- **Dynamically Verified UCR** ($N = 137$): Filtered by $T_{\text{dom}} \geq 16$ and $r_{\text{peak}} \geq 0.40$; guarantees macroscopic recurrence; rejects short-lag noise/trends.
- **Attractor & Grid:** Closed phase orbits $\implies \mathcal{G}_L = \{1, 2, 3, 4\} \times T_{\text{dom}}$.

2. Stochastic Random Walks & Jumps ($N = 64$ Series):

- **Domains:** Financial equities (Stock), clusters (Exathlon), telemetry (NEK, TAO).
- **Dynamical Profile:** Lack deterministic recurrence; monotonic ACF decay ($T_{\text{dom}} \rightarrow 1$); fail surrogate tests ($Z_{\text{det}} < 1.96$, linear stochastic/diffusion).
- **Routing & Grid:** Fraser-Swinney decorrelation $\implies \mathcal{G}_L = \{2^k \cdot \tau_{\text{AMI}}\}$.

5. Candidate Generation: Dual Horizon & Harmonic Dimension

1. Dual Horizon Selection ($L \in [8, N_{\text{tr}}/4]$):

- **Periodic** ($T_{\text{dom}} \geq 6$): $\mathcal{G}_L = \{1, 2, 3, 4\} \times T_{\text{dom}}$ (closed phase orbits).
- **Aperiodic** ($T_{\text{dom}} < 6$): $\mathcal{G}_L = \{2^k \cdot \tau_{\text{AMI}}\} \cap [8, N_{\text{tr}}/4]$ ($L \geq 8$ captures spikes).

2. Computing Harmonic Dimension ($d_{\text{harm}} = 2K + 1$):

- **FFT Peak Detection:** Finds K modes in $|X(f)|$ (prominence ≥ 0.08).
- **Quadrature Form:** K rotations + 1 trend $\implies d_{\text{harm}} = 2K + 1$.
- **Candidate Set:** $\mathcal{G}_d = \{1, 2, 4, 8\} \cup \{d_{\text{harm}}\}$ (bypasses eigengap collapse).

3. Classical Literature Grounding:

- **Broomhead & King (1986):** SSA trajectory rank for K modes is $2K + 1$.
- **Whitney-Takens (1991):** Invariant torus $\mathbb{T}^K$ embeds into $\mathbb{R}^{2K+1}$.

6. Ground-Truth Landscape: 4 Geometric Classes

Exhaustive Benchmark Oracle Audit (870 Datasets):

Sensitivity Class	Geometric Landscape	Count	Share
Class 1: Robust	Flat Pareto basin (scale & dim invariant)	211	24.3%
Class 2: Dim-Critical	Sensitive to d , invariant to horizon L	54	6.2%
Class 3: Horizon-Critical	Sensitive to L , robust across dimension d	348	40.0%
Class 4: Co-Sensitive	Joint (L, d) alignment mandatory	257	29.5%

Universal Invariant: $d^* \in \{1, 2, 4\}$ in $> 80\%$ of series, regardless of whether $L^* \in [15, 512]$.

7. Ground-Truth Anomaly Dynamics: Morphology & Contrast

Ground-Truth Audit (Evaluated at Oracle (L^* , d^*)):

Anom Type	Examples	Event Length	Peak Contrast	Max Z-Sep	Dynamical Driver
Point / Spikes	Stock, Yahoo	1–2	$3.2 \times -17.2 \times$	6.5σ	Small L (Energy $\propto 1/L$), high d
Subsequence	UCR, SED	$\approx T_{\text{dom}}$ (25–64)	$1.8 \times -2.5 \times$	3.6σ	$L \approx k \cdot T_{\text{dom}}$ (orbit closure)
Regime Shift	SMD, NEK, Exath.	$\gg T_{\text{dom}}$ (44–1800+)	$3.6 \times -118.6 \times$	2,464σ	Massive off-manifold excursion

Key Empirical Discoveries:

- **Boundary Overlap Dilution:** $\text{Contrast}_{\min} < 1.0 \times$ in **100%** of datasets ($[0.12 \times, 0.81 \times]$). Windows clipping edges contain 99% normal data; arithmetic averaging dilutes peak anomaly contrast by up to 86.5%.
- **Synthetic vs. Genuine Gap:** Synthetic training perturbations yield $\approx 1.15 \times$, whereas true anomalies exhibit $1.8 \times -118.6 \times$ peak orthogonal contrast.
- **Physical Justification for Apodization:** Motivates center-weighted Gaussian filtering to preserve sharp anomaly energy at window centers.

8. Determinism Audit: Surrogate Data Testing

IAAFT Null Hypothesis: Linear Gaussian process with static monotonic observation.

Determinism Significance Metric:

$$Z_{\text{det}} = \frac{\langle CD_{\text{surr}} \rangle - CD_{\text{orig}}}{\sigma(CD_{\text{surr}})}$$

Regime	Criterion	Count	Share	Mean Z
Stochastic / Linear	$Z < 1.96$	616	76.0%	-1.69
Mixed / Weak Determinism	$1.96 \leq Z < 5.0$	141	17.4%	+3.18
Strong Determinism	$Z \geq 5.0$	54	6.7%	+8.30

Key Finding: 76.0% of benchmarks lack a deterministic attractor; they behave as stochastic diffusion or autoregressive processes.

9. Determinism Mechanics: Correlation Dimension & IAAFT

1. Correlation Dimension (CD , Grassberger-Procaccia 1983):

- **Scaling Metric:** $C(r) = \frac{2}{N(N-1)} \sum_{i < j} \Theta(r - \|w_i - w_j\|_2) \propto r^{CD}$ as $r \rightarrow 0$.
- **Physical Meaning:** Measures phase-space fractal dimension.
Deterministic limit cycles $\implies CD \approx 1-2$; stochastic random walks $\implies CD \rightarrow m$.

2. The Central Limit Theorem Trap of Naive FFT:

- Summing random Fourier phases forces a **strictly Gaussian** distribution.
- Real telemetry with heavy tails/spikes falsely rejects $H_0 \implies$ **spurious determinism**.

3. IAAFT Alternating Projections (Schreiber & Schmitz 1996):

- **Spectral Step:** Replaces Fourier amplitudes with $|X_k^{\text{orig}}|$ (preserves ACF).
- **Rank Step:** Rank-orders back to exact original values (preserves histogram).
- **Rigorous Null:** $Z_{\text{det}} > 1.96$ isolates **true deterministic dynamics**.

10. Theory: Stochastic Takens & Codimension Disconnect

Stochastic Delay Embedding (Stark 1999):

$$d\mathbf{x}_t = f(\mathbf{x}_t) dt + \sigma(\mathbf{x}_t) d\mathbf{w}_t \implies \mathbf{x}_t \simeq \text{Bundle}(\mathcal{L}_{\text{Fokker-Planck}})$$

CNF and Conditional Diffusion parameterize the Fokker-Planck drift-diffusion operator.

The “Unfolding vs. Discrimination” Disconnect:

$\min m_{\text{Takens}} \implies k = 1$ (Minimal embedding) $\implies$ **Fails AD**

$\max k \implies L - d \gg 1$ (Normal bundle capacity) $\implies$ **Required**

Conclusion: L is governed by anomaly duration; d is governed by the rank of normal dynamics.

11. Horizon L : Periodic Regime ($T_{\text{dom}} \geq 6$)

Autocorrelation Operator:

$$R_{xx}(\tau) = \frac{1}{(N_{\text{tr}} - \tau)\sigma^2} \sum_{t=1}^{N_{\text{tr}} - \tau} (x_t - \mu)(x_{t+\tau} - \mu)$$

Dominant Fundamental Period:

$$T_{\text{dom}} = \arg \max_{\tau \geq 6} R_{xx}(\tau), \quad \text{prominence} \geq 0.03$$

Physical Harmonic Resonance Grid:

$$\mathcal{G}_L^{\text{periodic}} = \{1 \cdot T_{\text{dom}}, 2 \cdot T_{\text{dom}}, 3 \cdot T_{\text{dom}}, 4 \cdot T_{\text{dom}}\}$$

Coverage: Governs **73.1%** (629/860) of benchmark datasets.

Mechanism: Enforces topological orbit closure without phase boundary cuts.

12. Horizon L : Aperiodic Regime via Fraser-Swinney AMI

Failure of Linear Autocorrelation: $R_{xx}(\tau)$ decays monotonically

$\implies T_{\text{dom}} = 1.$

Average Mutual Information (Fraser & Swinney 1986):

$$I(\tau) = \iint p(x_t, x_{t+\tau}) \log \frac{p(x_t, x_{t+\tau})}{p(x_t)p(x_{t+\tau})} dx_t dx_{t+\tau}$$

First Local Information Minimum:

$$\tau_{\text{AMI}} = \arg \min_{\tau} I(\tau) \quad (\text{Maximal Non-Redundant Dynamical Info})$$

Dyadic Multi-Scale Candidate Grid:

$$\mathcal{G}_L^{\text{aperiodic}} = \text{clip}(\{2^k \cdot \tau_{\text{AMI}} \mid k \in \{1, 2, 3, 4, 5, 6\}\}, 8, L_{\text{max}})$$

Empirical Audit (231 Aperiodic Series): 97.0% clean minima;
61.5%–90.0% Oracle L^* hit rate (IOPS: 90.0%, WSD: 75.8%).

13. Test-Time Pointwise Projection: Anti-Dilution

Boundary Dilution of Arithmetic Mean Pooling:

$$s_{\text{arith}}(t) = \frac{1}{R_t} \sum_{k=0}^{L-1} s(W_{t-k})[k] \implies \text{Dilutes sharp spikes by 86.5\%}$$

Center-Weighted Gaussian Apodization:

$$w[k] = \exp\left(-\frac{1}{2} \left(\frac{k - (L-1)/2}{\sigma}\right)^2\right), \quad \sigma = \frac{L}{4}$$
$$s_{\text{gauss}}(t) = \frac{\sum_k w[k] \cdot s(W_{t-k})[k]}{\sum_k w[k]}$$

Soft-Max Generalized L_p -Norm Pooling:

$$s_{L_p}(t) = \left(\frac{1}{R_t} \sum_{k=0}^{L-1} s(W_{t-k})[k]^p \right)^{1/p}, \quad p \geq 2$$

Result: 303_UCR VUS-PR improves $0.2856 \rightarrow 0.3103$ (+8.6% relative gain).

14. Open Question: Intrinsic Dimension Estimation (d)

The Power-Law Spectral Decay Fallacy:

$$\lambda_k \propto k^{-\gamma}, \quad \gamma > 1 \quad (\text{Induced by Temporal Autocorrelation})$$

Universal Eigengap Collapse:

$$\arg \max_k (\lambda_k - \lambda_{k+1}) \equiv 1 \quad (\text{Trend Dominance})$$

The Under/Over Dilemma:

$d = 1 \implies$ Forces 1D line; normal limit cycles leak into residual

$d \geq 8 \implies$ Tangent space absorbs noise; destroys anomaly contrast

Status: Horizon L resolved via Dual Grid; Dimension d left as an open theoretical question.

15. Two-Stage Coarse-to-Fine Empirical Pilot

Dataset	Class	Selected (L, d)	Oracle (L^*, d^*)	Recovery %
234_SED	Class 4	(94, 4)	(100, 2)	103.6%
811_Exathlon	Class 1	(333, 2)	(100, 2)	99.9%
001_NAB	Class 3	(115, 16)	(70, 14)	99.2%
277_NEK	Class 2	(85, 16)	(15, 2)	94.8%
180_SMD	Class 2	(15, 4)	(100, 4)	93.2%
149_Stock	Class 1	(50, 4)	(15, 2)	81.2%
029_WSD (Static Grid)	Class 3	(333, 2)	(130, 2)	0.6%
029_WSD (AMI Grid)	Class 3	(110, 2)	(130, 2)	98.4%

Efficiency: Evaluates ≈ 35 – 45 models total vs. > 500 in full grid.

16. Summary & Roadmap

Completed Contributions:

- **Dual Grid Selection:** Unifies periodic ($k \cdot T_{\text{dom}}$) and aperiodic ($2^k \cdot \tau_{\text{AMI}}$) horizons without test leakage.
- **Direct Orthogonal Injection:** Strict off-manifold perturbation ($u_{\perp} \in \ker(V_d^{\top})$) for unsupervised contrast tuning.
- **Gaussian Apodization:** Suppresses boundary dilution by 86.5%.

Active Research Roadmap:

- **Dimension Estimation (d):** Investigate continuous Participation Ratio $d_{\text{eff}} = \frac{(\sum \lambda_i)^2}{\sum \lambda_i^2}$ and Grassmannian curvature to bypass power-law eigengap collapse.
- **Global Benchmark Run:** Execute dual coarse-to-fine dispatcher across all 860 benchmark datasets.